Mohamed A. Hanafy

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PROFESSIONAL SUMMARY

Researcher and Ph.D. Candidate in Electrical and Computer Engineering specializing in AI-driven brain-computer interfaces (BCI), multimodal signal processing, adaptive robotic control, assistive robotics, computer vision, and human-robot interaction (HRI). My NSF-supported research focuses on real-time decoding of human intent from EEG, physiological, behavioral, speech, and interaction signals to improve assistive mobility, rehabilitation robotics, patient-assistance systems, intelligent robotic platforms, industrial automation, and advanced manufacturing. My record includes peer-reviewed journal and conference publications, research conducted in connection with NSF-supported initiatives, reviewer service for international journals and conferences, leadership of multidisciplinary student research teams, competitive awards, and a concrete path to continue this work through ongoing and anticipated LARRI research projects with real-world practical applicability beyond academic research environments.

EDUCATION

Ph.D. Candidate in Electrical and Computer Engineering *University of Louisville, USA | 2022-Present*

- GPA: 3.89/4.00. Research focus: AI-enabled BCI, multimodal signal processing, adaptive control, intelligent assistive robotics, human-robot interaction, computer vision, and intelligent healthcare systems.
- Research conducted in connection with NSF-supported projects and University of Louisville research initiatives involving assistive robotics, advanced manufacturing, and healthcare-oriented human-machine interaction.

M.S. in Electronics and Communications Engineering *Zagazig University, Egypt | 2015 - 2019*

- Thesis: Compressed Sensing for Energy-Efficient Wireless Transmission.
- Research focus: signal processing, compressive sensing, brain-computer interfaces, and wireless healthcare telemonitoring.

B.S. in Electronics and Communications Engineering *Zagazig University, Egypt | 2008 - 2013*

- Graduation Project: Wireless Sensor Body Area Network (WSBAN). Excellent with honors; ranked 1st in undergraduate class.

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant — AI, BCI, Robotics, and Human-Robot Interaction *Sep. 2023 - Present*

Louisville Automation & Robotics Research Institute (LARRI), University of Louisville, USA

- **Brain-Computer Interface Frameworks:** Design, optimize, and implement EEG-based BCI frameworks for real-time motor-intent decoding, human-robot interaction, and assistive healthcare and rehabilitation systems by improving data quality, response time, and classification accuracy.
- **Neural and Adaptive Control Systems:** Develop AI-based adaptive robot teleoperation interfaces and neuroadaptive control approaches that improve interaction smoothness, responsiveness, usability, efficiency, adaptability, and safety for assistive and human-centered robotic platforms.
- **ARNA and Healthcare Robotics:** Contribute to Adaptive Robotic Nursing Assistant (ARNA) research involving patient ambulation support, tactile sensing, adverse-event detection, physical human-robot interaction, and clinically oriented robotic assistance workflows.
- **Computer Vision and Logistics:** Implement and optimize computer vision and AI methods to enhance object detection, grasp efficiency, and barcode identification for real-time robotic perception and precise tracking in industrial automation and UPS logistics research environments.
- **NSF Study Coordination and Compliance:** Appointed Study Coordinator for NSF-supported KAMPERS/KY NSF EPSCoR research (1849213) activities through University of Louisville IRB documentation, supporting research execution, active-study coordination, study protocols, and compliance responsibilities.
- **Project Integration:** Contribute to NSF-supported and industry-supported LARRI research involving assistive robotics, adaptive control, BCI systems, healthcare-oriented human-robot interaction, and industrial automation.
- **Leadership and Dissemination:** Lead and mentor multidisciplinary student researchers in BCI research, neural data acquisition, experiment design, EEG processing, model evaluation, and performance analysis; authored and co-authored peer-reviewed papers disseminating research outcomes.

Research Collaborator — Multimodal Speech and Healthcare Analytics *Jan. 2026 - Present*

University of Louisville (LARRI; Psychological & Brain Sciences, Dr. Maria Kondaurova) and Washington State University (Speech and Language Lab, College of Medicine, Dr. Mark VanDam)

- Collaborate on a longitudinal speech and language research project involving pediatric communication, child language development, hearing loss, and healthcare analytics by providing technical expertise in programming, signal processing, and machine-learning workflows for healthcare-oriented speech-analysis applications.

- Develop automated acoustic-analysis pipelines to extract, organize, and analyze speech and audio features from large naturalistic recordings, extending multimodal signal-processing expertise beyond neural signals to speech and communication data.
- Selected as Graduate Research Assistant and Study Coordinator for the Jon Rieger Seed Grant project, with responsibilities involving speech emotion recognition algorithms, study design/IRB-related protocols, and supervision of undergraduate research teams.

Graduate Research Assistant — Computer Vision & Assistive Technologies *Jan. 2022 - Aug. 2023*

Computer Vision and Image Processing (CVIP) Laboratory, University of Louisville, USA

- Conducted applied research in machine learning, digital image processing, computer vision, and assistive technologies.
- Contributed to a visual assistance system for visually impaired individuals, including obstacle detection, object recognition, and scene-description systems.
- Utilized Neural Radiance Fields (NeRFs) to generate 3D street-level mapping from monocular video inputs, supporting autonomous perception systems.

Graduate Teaching Assistant *2023 - 2026*

Department of Electrical and Computer Engineering, University of Louisville, USA

- Instructed and supported laboratory courses in control systems, digital signal processing, embedded systems, advanced microprocessors, and electrical/computer engineering fundamentals.
- Supported and redesigned laboratory materials, grading rubrics, experiments, assignments, and student technical guidance for undergraduate and graduate engineering courses.
- Delivered lectures and mentored students in signal processing, control systems, circuit analysis, programming, and robotics-related laboratories.

Graduate Research and Teaching Assistant — Signal Processing & Healthcare Telemonitoring *2015 - 2021*

Department of Electronics and Communications Engineering, Zagazig University, Egypt

- Project Lead: Developed advanced compressed sensing and block sparse Bayesian learning methods for low-power, high-reliability wireless transmission over body area networks, establishing a signal-processing foundation for later BCI and neural-decoding research.
- Instruction & Mentorship: Taught laboratory and tutorial sessions for signal/image processing, analog and digital communications, logic circuit design, electronic circuits and devices, wave propagation, and numerical analysis.
- Engineering Leadership: Supervised undergraduate capstone projects involving hands-on technical skills, system-level design, and engineering innovation.

FUNDED RESEARCH PROJECTS AND INNOVATION

- Conducted research in connection with **NSF EPSCoR RII Track-1 / KAMPERS Award #1849213**, "Kentucky Advanced Manufacturing Partnership for Enhanced Robotics and Structures," (2019–2024; UofL subcontract budget: $6,150,000; UofL PI: Dr. Dan O. Popa; lead PI: Dr. Rodney Andrews, University of Kentucky). My work contributed to BCI, adaptive robotic control, human-robot interaction, and advanced robotic systems within the LARRI research environment.
- Conducted research in connection with **NSF Award #2026584**, "Enhancing Future Work of Nursing Professionals through Collaborative Human-Robot Interfaces," a $1.5 million NSF Future of Work project led by Dr. Dan O. Popa at LARRI, University of Louisville. My work contributed to collaborative human-robot interfaces and the Adaptive Robotic Nursing Assistant (ARNA) platform.
- Contributed to **UPS-supported industrial automation research**, "Robotic Manipulation for Sorting Exception Area Phase I and Phase II" (2023–2025; approximately $791K). My work focuses on robotic manipulation, object detection, grasp efficiency, barcode identification, and logistics automation within LARRI.
- Research collaboration supported by the **Jon Rieger Seed Grant**, "AI-enabled analysis of emotions in daylong child-centered speech recordings" (Spring 2026 award; award period July 1, 2026–June 30, 2027; PI: Dr. Maria Kondaurova; co-PIs: Dr. Dan O. Popa and Dr. Mark VanDam). My work contributed to machine-learning, active-learning, annotation, and acoustic-analysis workflows for speech emotion recognition and child-centered speech recordings.
- LaunchIt Innovation Bootcamp / **NSF-supported regional I-Corps Hub** program, 2025 - completed customer discovery, market validation, and research-to-market training for the Mind Pilot BCI/assistive robotics concept.

PUBLICATIONS AND MANUSCRIPTS [Google Scholar]

Journal Articles

1. **M. A. Hanafy**, Saykhun Yusufjonov, P. SharafianArdakani, D. Yusufjonov, M. M. Rayguru, D. O. Popa, "A Dual-Validation Framework for Temporal Robustness Assessment in Motor Imagery Brain-Computer Interfaces, "Technologies, MDPI, 2025.
2. A. Prakash, **M. A. Hanafy**, J. Dowdy, D. O. Popa. "Real-Time Neuroadaptive Control with Tactile Calibration for Physical Human-Robot Interaction", Electronics, MDPI, 2026.
3. **M. A. Hanafy**, H. S. Ali, A. A. Shaalan, "Rakeness with Block Sparse Bayesian Learning for Efficient ZigBee-Based EEG Telemonitoring", International Journal of Communication Systems, Wiley, 2019.
4. P. Sharafian Ardakani, S. Abubakar, S. K. Das, S. Surupa, **M. A. Hanafy**, M. Rayguru, and D. O. Popa, "Evaluation of a Neuroadaptive Controller in Admittance Control of a Mobile Robot for Physical Human-Robot Interaction", Intelligent Service Robotics, 2025.
5. P. SharafianArdakani, **M. A. Hanafy**, I. Kondaurova, A. Ashary, M. M. Rayguru, and D. O. Popa, "Adaptive User Interface with Parallel Neural Networks for Robot Teleoperation", IEEE Robotics and Automation Letters (RA-L), 2024; presented at ICRA 2025.

Conference Papers

6. **M. A. Hanafy**, A. Prakash, S. Yusufjonov, M. M. Rayguru, D. O. Popa, "Toward Robust Assistive Robot Navigation with EEG Motor-Imagery, IEEE IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) workshop, 2026.
7. **M. A. Hanafy**, S. Yusufjonov, D. Yusufjonov, P. Sharafian Ardakani, M. M. Rayguru, and D. O. Popa, "A Robust EEG Brain-Computer Interface Approach for Decoding Hand and Foot Motor Imagery", Proceedings of the ACM 18th International Conference on PErvasive Technologies Related to Assistive Environments (PETRA), 2025.
8. **M. A. Hanafy**, H. L. Reynolds, J. Jones, D. Harris, D. O. Popa, "Assessment of BCI Performance for Human-Robot Interaction", Proceedings of the ACM 17th International Conference on PErvasive Technologies Related to Assistive Environments (PETRA), 2024.
9. **M. A. Hanafy**, H. S. Ali, A. A. Shaalan, "Compressed Sensing for Reliable Body Area Propagation with Efficient Signal Reconstruction", Proceedings of the 29th International Conference on Microelectronics (ICM), 2017.
10. H. L. Reynolds, **M. A. Hanafy**, J. Jones, D. Harris, D. O. Popa, "Comparative Study of One-Handed vs. Two-Handed EEG Intent Recognition for Applications in Human-Robot Interaction", Proceedings of the ACM 17th International Conference on PErvasive Technologies Related to Assistive Environments (PETRA), 2024**. Best Workshop Student Paper Award**

Manuscripts Under Review

11. Tactile Handlebar and Deep Learning for Safe Interaction with a Robot Nursing Assistant, submitted to IEEE Transactions on Medical Robotics and Bionics, 2026.

REVIEWER ROLES AND PROFESSIONAL SERVICE

- Reviewer, IEEE Transactions on Systems, Man, and Cybernetics: Systems.
- Reviewer, IEEE Journal of Biomedical and Health Informatics (JBHI).
- Reviewer, IEEE Access.
- Reviewer, International Journal of Digital Health (JDH).
- Reviewer, Frontiers in Robotics and AI.
- Reviewer, ACM PETRA Conference.
- Reviewer, IEEE SoutheastCon.
- Reviewer, IEEE International Conference of Smart Mobility (IEEE SM).
- Reviewer, IEEE International Conference on Control, Automation and Instrumentation (IEEE IC2AI).
- International Program Committee Member, 2025 International Symposium on Embodied Intelligence and Humanoid Robots (Roboi).
- Judge for robotics competitions, including Redbird Robotics and the VEX V5RC Kentucky State Championship.

AWARDS AND HONORS

- Theobald Scholarship Award, University of Louisville, 2025 - awarded for academic achievement, entrepreneurial spirit, civic involvement, and commitment to engineering science.
- NSF PETRA Conference Doctoral Consortium Award, ACM PETRA Conference, 2024.
- Best Workshop Student Paper Award, ACM PETRA Conference, 2024.
- Outstanding Poster Presentation Award, KY NSF EPSCoR, 2024.
- COABE Students as Leaders Mini Grant, 2025.
- Degree of Honor of Cumulative Academic Distinction; ranked 1st in undergraduate class, Zagazig University, 2013.
- College Excellence Award, awarded to distinctive students during all five undergraduate years, Zagazig University, 2008-2013.

SELECTED PRESENTATIONS, POSTERS, and RESEARCH DISSEMINATION

1. Evaluation of a Robotic Nursing Assistant from Neuroadaptive Ambulation to Hospital Deployment, IEEE IROS, Pittsburgh, 2026.
2. Controlling a Robotic Arm with Brain Activity via a Binary Classifier, SUPERCOLLIDER, Lexington, 2024.
3. Personalized Adaptive User Interfaces for Robot Teleoperation, SUPERCOLLIDER, Lexington, 2024.
4. Conference participation and research presentation activities include IEEE IROS 2026, ACM PETRA 2024, ACM PETRA 2025, IEEE SoutheastCon 2026, IEEE ICRA 2025, and IEEE ICM 2017.

PROFESSIONAL DEVELOPMENT

- G.tech BCI & Neurotechnology Spring School, 2026 - completed 130 hours of advanced neurotechnology and BCI training; also invited to attend the 2026 G.tech Conference.
- Graduate Publishing Academy Certificate, University of Louisville, 2025.
- Graduate Teaching Academy Certificate, University of Louisville, 2025.
- Graduate Community Engaged Scholarship Certificate, University of Louisville, 2025.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB & Simulink, C++, VHDL.
- **AI & ML:** TensorFlow, Keras, Scikit-learn.
- **Computer Vision:** Object detection and recognition (YOLO, semantic segmentation), 3D vision, multi-view geometry, image processing (OpenCV, PIL).
- **Data Analysis & Visualization:** NumPy, Pandas, SciPy, Matplotlib, Plotly.
- **Hardware:** EEG systems, sensor fusion, Arduino, Raspberry Pi, STM32, Proteus.
- **Robotics Platforms & Simulation:** ROS/ROS2, Gazebo, Isaac Sim.
- **Tools & OS:** Git, GitHub, Linux, Windows.

SELECTED RESEARCH CODE AND TECHNICAL PROJECTS

- Maintain a GitHub portfolio (github.com/MAHanafy) documenting selected research-related code, software workflows, and implementation.
- Brain-Computer Interface for Assistive Robotics: Led and mentored a student research team developing EEG-based BCI workflows for real-time robotic control and assistive human-robot interaction.
- Neural Networks for Robot Teleoperation: Developed adaptive user-interface and parallel neural-network methods to improve robot teleoperation performance and human-robot collaboration.
- Industrial Automation and Logistics Robotics: Implemented computer-vision algorithms for object detection, grasp-efficiency analysis, barcode identification, and robotic perception in logistics-oriented automation environments.
- Longitudinal Speech and Hearing Research: Developed software workflows to analyze acoustic features in child-centered recordings, including pitch, speech clarity, and distinctions between child-directed and adult-directed speech.